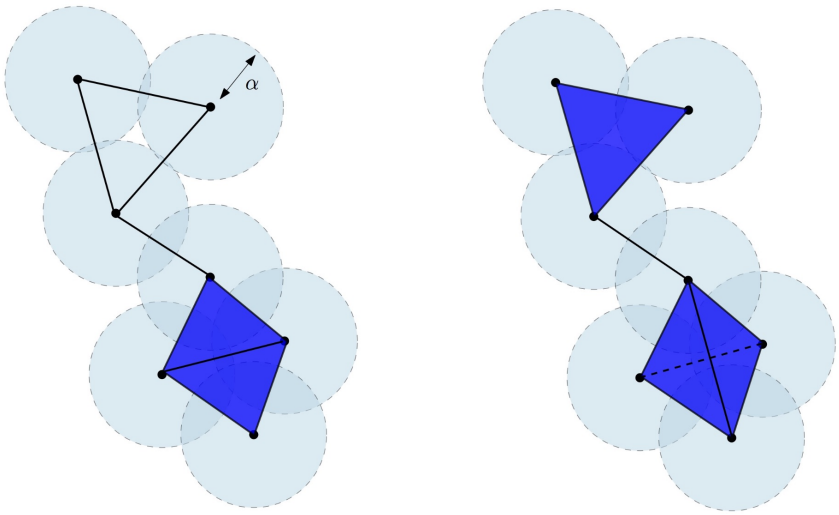

Figure 8: The effect of outliers on the sublevel sets of distance functions. Adding just a few outliers to a point cloud may dramatically change its distance function and the topology of its offsets.

Given a probability distribution P in $\mathbb{R}^d$ and a real parameter $0 \leq u \leq 1$, the notion of distance to the support of P may be generalized as the function

$$\delta_{P,u} : x \in \mathbb{R}^d \mapsto \inf\{t > 0\,;\, P(B(x,t)) \geq u\},$$

where $B(x,t)$ is the closed Euclidean ball of center x and radius t. To avoid issues due to discontinuities of the map $P \to \delta_{P,u}$, the distance-to-measure function (DTM) with parameter $m \in [0,1]$ and power $r \geq 1$ is defined by

$$d_{P,m,r}(x) : x \in \mathbb{R}^d \mapsto \left(\frac{1}{m}\int_0^m \delta_{P,u}^r(x)du\right)^{1/r}. \tag{1}$$

A nice property of the DTM proved in Chazal et al. (2011b) is its stability with respect to perturbations of P in the Wasserstein metric. More precisely, the map $P \to d_{P,m,r}$ is $m^{-\frac{1}{r}}$-Lipschitz, i.e. if P and $\tilde{P}$ are two probability distributions on $\mathbb{R}^d$, then

$$\|d_{P,m,r} - d_{\tilde{P},m,r}\|_\infty \leq m^{-\frac{1}{r}} W_r(P,\tilde{P}) \tag{2}$$

where W_r is the Wasserstein distance for the Euclidean metric on $\mathbb{R}^d$, with exponent r[9]. This property implies that the DTM associated to close distributions in the Wasserstein metric have close sublevel sets. Moreover, when $r = 2$, the function $d^2_{P,m,2}$ is semiconcave ensuring strong regularity properties on the geometry of its sublevel sets. Using these properties, Chazal et al. (2011b) show that, under general assumptions, if $\tilde{P}$ is a probability distribution approximating P, then the sublevel sets of $d_{\tilde{P},m,2}$ provide a topologically correct approximation of the support of P.

In practice, the measure P is usually only known through a finite set of observations $\mathbb{X}_n = \{X_1, \ldots, X_n\}$ sampled from P, raising the question of the approximation of the DTM. A natural idea to estimate the DTM from $\mathbb{X}_n$ is to plug the empirical measure P_n instead of P in the definition of the DTM. This "plug-in strategy" corresponds to computing the distance to the

[9]See Villani (2003) for a definition of the Wasserstein distance

empirical measure (DTEM). For $m = \frac{k}{n}$, the DTEM satisfies

$$d^r_{P_n,k/n,r}(x) := \frac{1}{k}\sum_{j=1}^{k} \|x - \mathbb{X}_n\|^r_{(j)} ,$$

where $\|x - \mathbb{X}_n\|_{(j)}$ denotes the distance between x and its j-th neighbor in $\{X_1, \ldots, X_n\}$. This quantity can be easily computed in practice since it only requires the distances between x and the sample points. The convergence of the DTEM to the DTM has been studied in Chazal et al. (2014a) and Chazal et al. (2016b).

The introduction of DTM has motivated further works and applications in various directions such as topological data analysis (Buchet et al., 2015a), GPS traces analysis (Chazal et al., 2011a), density estimation (Biau et al., 2011), hypothesis testing Brécheteau et al. (2019), clustering (Chazal et al., 2013b) just to name a few. Approximations, generalizations and variants of the DTM have also been considered in (Brécheteau et al., 2020; Buchet et al., 2015b; Guibas et al., 2013; Phillips et al., 2014).

5 Persistent homology

Persistent homology is a powerful tool to compute, study and encode efficiently multiscale topological features of nested families of simplicial complexes and topological spaces. It does not only provide efficient algorithms to compute the Betti numbers of each complex in the considered families, as required for homology inference in the previous section, but also encodes the evolution of the homology groups of the nested complexes across the scales. Ideas and preliminary results underlying persistent homology theory trace back to the 20th century, in particular in Barannikov (1994); Frosini (1992); Robins (1999). It started to know an important development in its modern form after the seminal works of Edelsbrunner et al. (2002) and Zomorodian and Carlsson (2005).

5.1 Filtrations

A *filtration of a simplicial complex* K is a nested family of subcomplexes $(K_r)_{r\in T}$, where $T \subseteq \mathbb{R}$, such that for any $r, r' \in T$, if $r \leq r'$ then $K_r \subseteq K_{r'}$, and $K = \cup_{r\in T} K_r$. The subset T may be either finite or infinite. More generally, a *filtration* of a topological space $\mathbb{M}$ is a nested family of subspaces $(M_r)_{r\in T}$, where $T \subseteq \mathbb{R}$, such that for any $r, r' \in T$, if $r \leq r'$ then $M_r \subseteq M_{r'}$ and, $M = \cup_{r\in T} M_r$. For example, if $f : \mathbb{M} \to \mathbb{R}$ is a function, then the family $M_r = f^{-1}((-\infty, r])$, $r \in \mathbb{R}$ defines a filtration called the sublevel set filtration of f.

In practical situations, the parameter $r \in T$ can often be interpreted as a scale parameter and filtrations classically used in TDA often belong to one of the two following families.

Filtrations built on top of data. Given a subset $\mathbb{X}$ of a compact metric space (M, ρ), the families of Rips-Vietoris complexes $(\mathrm{Rips}_r(\mathbb{X}))_{r\in\mathbb{R}}$ and and Čech complexes $(\mathrm{Cech}_r(\mathbb{X}))_{r\in\mathbb{R}}$ are filtrations [10]. Here, the parameter r can be interpreted as a resolution at which one considers the data set $\mathbb{X}$. For example, if $\mathbb{X}$ is a point cloud in $\mathbb{R}^d$, thanks to the Nerve theorem, the filtration $(\mathrm{Cech}_r(\mathbb{X}))_{r\in\mathbb{R}}$ encodes the topology of the whole family of unions of balls $\mathbb{X}^r = \cup_{x\in\mathbb{X}} B(x, r)$, as r goes from 0 to $+\infty$. As the notion of filtration is quite flexible, many other filtrations have been considered in the literature and can be constructed on top of data, such as e.g. the so called witness complex popularized in TDA by De Silva and Carlsson (2004), the weighted Rips filtrations Buchet et al. (2015b), or the so-called DTM-filtrations Anai et al. (2020) that allow to handle data corrupted by noise and outliers.

[10] we take here the convention that for $r < 0$, $\mathrm{Rips}_r(\mathbb{X}) = \mathrm{Cech}_r(\mathbb{X}) = \emptyset$

Sublevel sets filtrations. Functions defined on the vertices of a simplicial complex give rise to another important example of filtration: let K be a simplicial complex with vertex set V and $f : V \to \mathbb{R}$. Then f can be extended to all simplices of K by $f([v_0, \cdots, v_k]) = \max\{f(v_i) : i = 1, \cdots, k\}$ for any simplex $\sigma = [v_0, \cdots, v_k] \in K$ and the family of subcomplexes $K_r = \{\sigma \in K : f(\sigma) \leq r\}$ defines a filtration call the sublevel set filtration of f. Similarly, one can define the upperlevel set filtration of f.

In practice, even if the index set is infinite, all the considered filtrations are built on finite sets and are indeed finite. For example, when $\mathbb{X}$ is finite, the Vietoris-Rips complex $\mathrm{Rips}_r(\mathbb{X})$ changes only at a finite number of indices r. This allows to easily handle them from an algorithmic perspective.

5.2 Starting with a few examples

Given a filtration $\mathrm{Filt} = (F_r)_{r \in T}$ of a simplicial complex or a topological space, the homology of F_r changes as r increases: new connected components can appear, existing component can merge, loops and cavities can appear or be filled, etc... Persistent homology tracks these changes, identifies the appearing features and associates a life time to them. The resulting information is encoded as a set of intervals called a *barcode* or, equivalently, as a multiset of points in $\mathbb{R}^2$ where the coordinate of each point is the starting and end point of the corresponding interval.

Before giving formal definitions, we introduce and illustrate persistent homology on a few simple examples.

Example 1. Let $f : [0, 1] \to \mathbb{R}$ be the function of Figure 9(a) and let $(F_r = f^{-1}((-\infty, r]))_{r \in \mathbb{R}}$ be the sublevel set filtration of f. All the sublevel sets of f are either empty or a union of intervals, so the only non trivial topological information they carry is their 0-dimensional homology, i.e. their number of connected components. For $r < a_1$, F_r is empty, but at $r = a_1$ a first connected component appears in F_{a_1}. Persistent homology thus registers a_1 as the birth time of a connected component and start to keep track of it by creating an interval starting at a_1. Then, F_r remains connected until r reaches the value a_2 where a second connected component appears. Persistent homology starts to keep track of this new connected component by creating a second interval starting at a_2. Similarly, when r reaches a_3, a new connected component appears and persistent homology creates a new interval starting at a_3. When r reaches a_4, the two connected components created at a_1 and a_3 merges together to give a single larger component. At this step, persistent homology follows the rule that this is the most recently appeared component in the filtration that dies: the interval started at a_3 is thus ended at a_4 and a first persistence interval encoding the lifespan of the component born at a_3 is created. When r reaches a_5, as in the previous case, the component born at a_2 dies and the persistent interval (a_2, a_5) is created. The interval created at a_1 remains until the end of the filtration giving rise to the persistent interval (a_1, a_6) if the filtration is stopped at a_6, or $(a_1, +\infty)$ if r goes to $+\infty$ (notice that in this later case, the filtration remains constant for $r > a_6$). The obtained set of intervals encoding the span life of the different homological features encountered along the filtration is called the *persistence barcode* of f. Each interval (a, a') can be represented by the point of coordinates (a, a') in $\mathbb{R}^2$ plane. The resulting set of points is called the *persistence diagram* of f. Notice that a function may have several copies of the same interval in its persistence barcode. As a consequence, the persistence diagram of f is indeed a multi-set where each point has an integer valued multiplicity. Last, for technical reasons that will become clear in the next section, one adds to the persistence all the points of the diagonal $\Delta = \{(b, d) : b = d\}$ with an infinite multiplicity.

Example 2. Let now $f : M \to \mathbb{R}$ be the function of Figure 9(b) where M is a 2-dimensional surface homeomorphic to a torus, and let $(F_r = f^{-1}((-\infty, r]))_{r \in \mathbb{R}}$ be the sublevel set filtration of f. The 0-dimensional persistent homology is computed as in the previous example, giving rise to the red bars in the barcode. Now, the sublevel sets also carry 1-dimensional homological